

Sequence detector

A sequence detector is an FSM that watches a serial bit stream and raises Z when a target pattern appears

Detect 1011 starter

- Starter: Mealy overlapping detector for pattern one-zero-one-one
- Step stream one-zero-one-one, Z goes zero, zero, zero, one on the last bit
- States track matched prefix: empty, one, one-zero, one-zero-one
- Toggle overlap off and see restart behavior
- Switch to Moore and Z comes from the detect state instead of the arc
- Try pattern one-one-zero-one or a longer stream with two overlapping matches

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Sequence detector playground

Step a bit stream through a small FSM that detects a binary pattern (Mealy or Moore, overlapping or not). Watch state, next-state, and output Z. Starter: detect 1011 (Mealy, overlap).

Starter example: Mealy overlapping detector for 1011. Step the stream 1011 — Z goes 0,0,0,1.

Challenge 0 / 22 cleared

Quiz: detector: A sequence detector FSM recognizes...

☐ a specific bit pattern in a serial input stream

☐ only combinational AND

☐ DRAM refresh timing

☐ I²C address only

Show hint

Check

Next

Load challenge setup

Idle

Quiz: detector

Quiz: Mealy

Quiz: Moore

Quiz: overlap

Quiz: states

Quiz: Z=1

Quiz: serial

Quiz: non-overlap

Detect 1011

After 10

Z on last bit

No false detect

Overlap 1011

Moore DETECT

Pattern 1101

Reset to S0

Sequence detector starter

Workbook practice

- In the workbook track
- Fill the transition table for S2 on inputs zero and one
- Explain what state you enter after a failed bit
- Sketch one overlapping case: one-zero-one-one-one should match twice
- Name one pitfall: forgetting overlap when the pattern repeats inside itself

Pitfalls to watch

- Do not confuse Mealy pulse timing with Moore level timing, downstream logic may care
- Prefix recovery is not optional; brute-force reset wastes matches
- Non-overlap mode is a different spec, know which you need
- And remember: the browser lab is literacy
- Real RTL still needs correct encoding, reset state, and test streams

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one Mealy detect arc and one overlap restart
- When you are ready, take the short quiz, then continue to ring and Johnson counters